FINETUNING GRAPH HYPERNETWORK USING LoRA (Low Rank Adaptation)

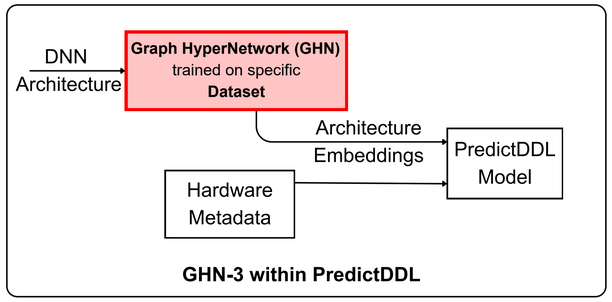
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Introduction

- Predicting Deep Neural Network (DNN) training time enables efficient utilization of computational resources, reducing cost and accelerating model development.
- PredictDDL uses Graph HyperNetwork (GHN) to predict DNN training time without fully training the model.
- Limitation: Adapting GHN to a new dataset traditionally requires full retraining.

Background

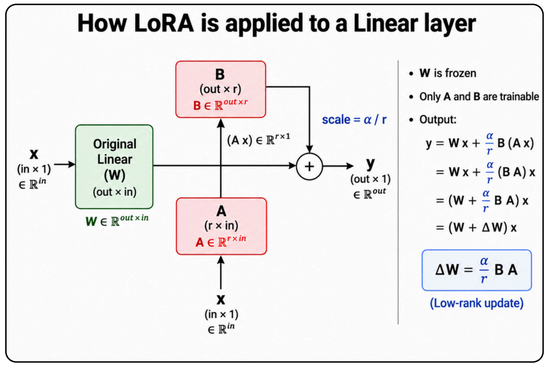


- GHN-3 predicts the weights of a target neural network directly from its architecture
- GHN-3 is pretrained on ImageNet. Adapting it to a new dataset traditionally requires retraining the model.

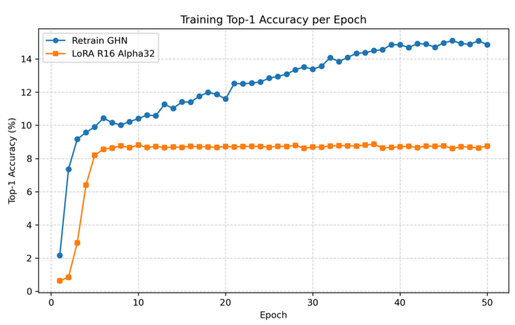
Research Question:
Can LoRA make GHN-3 adaptation to new dataset more efficient without significantly degrading predictive performance?

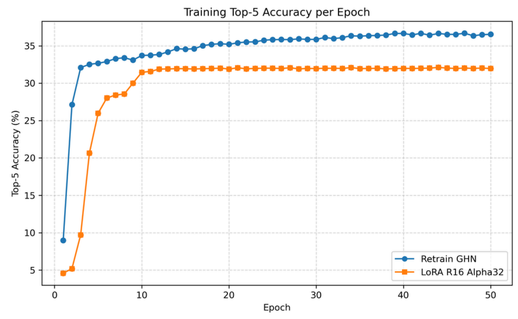
Low Rank Adaptation (LoRA) Integration

- LoRA is a parameter efficient fine-tuning technique.
- The pretrained weights are frozen, while low-rank matrices **A** and **B** are learned.
- Extended GHN-3 to support the Caltech-101 dataset.
- Replaced Linear layers in the GHN-3 architecture with LoRA Linear layers.
- Initialized from ImageNet-pretrained GHN-3 weights.
- Fine-tuned only the LoRA parameters.



Results

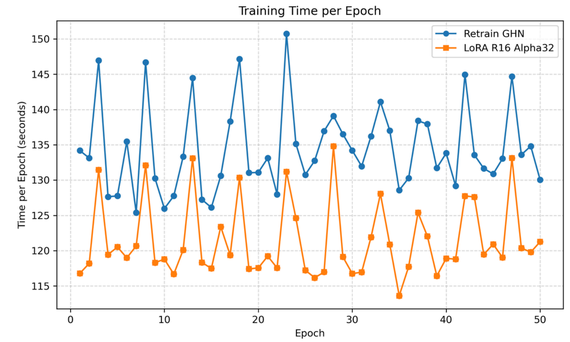




- Compared a fully retrained GHN-3 on Caltech-101 dataset with a GHN-3 model with ImageNet weights finetuned using LoRA for Caltech-101 dataset.

Hyperparameters	Value
Learning Rate	$5e^{-5}$
Meta Batch Size	8
Batch Size	32

Hyperparameter	Value
LoRA Rank	16
LoRA Alpha	32
LoRA Dropout	0.05



	Retrained Model	LoRA Finetuned Model
Total Parameters	7,986,052	7,986,052
Trainable Parameters	7,986,052 (100%)	**1,209,420 (15.14%)**

Key Takeaways

- LoRA enables adaptation of an ImageNet-pretrained GHN-3 to the Caltech-101 dataset.
- LoRA significantly reduces the computational cost of adapting GHN-3 in terms of memory and training time.
- Full retraining achieved higher accuracy, highlighting the trade-off between computational efficiency and predictive performance.

Future Work

- Evaluate the approach on additional datasets.
- Evaluate the approach by applying LoRA to different parts of GHN-3

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References

[1] E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," ICLR, 2022.
[2] B. Knyazev, D. Hwang, and S. Lacoste-Julien, "Can We Scale Transformers to Predict Parameters of Diverse ImageNet Models?," arXiv:2303.04143, 2023.
[3] K. Assogba, E. Lima, M. M. Rafique, and M. Kwon, "PredictDDL: Reusable Workload Performance Prediction for Distributed Deep Learning," IEEE CLUSTER, 2023.